$(pos, 2i)$ $\begin{cases} 2i = 0 \Rightarrow \sin \Rightarrow -0.756 + 0.1 = \\ 2i+1 = 1 \Rightarrow \cos \Rightarrow -0.6536 + 0 = \end{cases}$ $i = 0$

$i = 1 \begin{cases} 2 \\ 3 \end{cases}$ $+ 0.23 =$

$i = 2 \begin{cases} 4 \\ 5 \end{cases}$

add them up.

LLM Week 3

Graded Assignment - 3

1) Consider the embedding vector for a word, $x = [0.1, 0, 0.23, 0.4, -0.75, 1]$. Suppose the word is at position 4 in the given sentence. Add the corresponding position embedding p to the word embedding and enter the sum of the elements in $h = x + p$.Use the fixed-sinusoidal position embedding vector calculated using the formula given below

$PE(pos, 2i) = sin\left(\frac{pos}{10000^{2i/dmodel}}\right)$

$PE(pos, 2i+1) = cos\left(\frac{pos}{10000^{2i/dmodel}}\right)$

len $= 6$, $d_{model} = 6$

pos $= 4$

$i = 0, 1, 2$

now once you get $p \Rightarrow h = x + p$

1.75

Yes, the answer is correct.
Score: 2

Accepted Answers:

(Type: Range) 1.7,1.8

2 points

2) Consider a language where sentences are written from Right-to-Left (RTL) direction instead of Left-to-Right writing (reading) method. An example sentence is ``transformer movie the enjoyed I". Suppose we decided to use the GPT for Causal Language Modelling (CLM) of the text, then *1 point*

- It is necessary to flip the attention mask horizontally
- It is necessary to flip the attention mask vertically
- Neither the horizontal nor the vertical flipping of attention mask is required

CLM works on the principle that at pos t, the model is allowed to see token at pos $\leq t$.

$\therefore$, it doesn't matter

Yes, the answer is correct.
Score: 1

Accepted Answers:

Neither the horizontal nor the vertical flipping of attention mask is required

3) Choose the correct statements *1 point*

- All the parameters of the GPT model are randomly initialized during pre-training
- The model minimizes the CLM objective during fine-tuning to improve the performance
- All parameters of the GPT model are randomly initialized during fine-tuning
- In general, fine-tuning requires a dataset with labels

Yes, the answer is correct.
Score: 1

Accepted Answers:

All the parameters of the GPT model are randomly initialized during pre-training
In general, fine-tuning requires a dataset with labels

4) Assume that we have a large corpus of text. The vocabulary constructed from the text contains 10000 words. Of these, 100 words occurred only once in the entire corpus of text. The parameters of the embedding layer and the output layer of the model are shared. Suppose we create a batch of 256 samples (each sample is a sentence from the corpus). None of these samples contains any of the 100 rare words. Suppose we pre-train the model for one iteration using the batch of samples, then *2 points*

- it is certain that the embeddings of none of these 100 rare words will get updated.
- there is a chance that the embeddings of all or some of these 100 rare words will get updated
- the embeddings of all these 100 rare words will defintely get updated

Yes, the answer is correct.
Score: 2

Accepted Answers:

there is a chance that the embeddings of all or some of these 100 rare words will get updated

Rajesh has created a GPT-like transformer model. However he doesn't have access to large computation power, so he has configured the original architecture in the following manner:

1. He has taken a vocabulary of size 1000 words/tokens.
2. Sequence length is 64.
3. Embedding dimension and d_{model} is 128.
4. There are only 4 transformer blocks (layers).
5. Each transformer block (layer) has only 4 attention heads.
6. FFN hidden layer size is 256.

5) What will be the shape of positional embedding? **1 point**

→ encode each Token

- [x] 64×128
- [ ] 64×4
- [ ] 12×128
- [ ] 512×768

T_{PE} = seq. len × emb dim

= 64 × 128

Yes, the answer is correct.
Score: 1

Accepted Answers:

64×128

6) What will be the shape of W_Q^i? **1 point**

- [ ] 32×32
- [ ] 64×32
- [ ] 128×16
- [x] 128×32

$d_{model} = 128$
heads = 4

head dim = $\frac{128}{4}$ = 32

$W_Q = W_K = W_V$ = input dim × head dim

= 128 × 32

7) What will be shape of the mask M? **1 point**

- [ ] 16×16
- [ ] 32×32
- [x] 64×64
- [ ] 128×128
- [ ] None of these.

$[\ A\ ] + [\ \ ]_M =$

T × T → always seq × seq len

Yes, the answer is correct.
Score: 1

Accepted Answers:

64×64

8) Consider the following set of GPT-based text generative models (represented by M_i with $i \in \{1, 2, 3, 4\}$) and the corresponding text generated with the prompt "I like" **1 point**

M_1 : I like to to like.
M_2 : I like like like.
M_3 : I like the do because.
M_4 : I like to study the origin of the universe.
Select the model(s) that is(are) degenerative.

→ repetiti", loop, nonsensical statements

- [x] M_1
- [x] M_2
- [x] M_3
- [ ] M_4